

Using AI/ML to Predict School Attendance AI/ML

The problem I want to address is identifying students who are at risk of poor attendance and potential dropout. School attendance is often a strong indicator of whether a student will successfully graduate (RAND Corporation, 2025). Students who frequently miss school are more likely to fall behind academically, develop behavioral issues, or eventually leave school entirely. Some students who struggle with attendance may also be dealing with challenges outside of school, such as unstable housing, family responsibilities, or a lack of transportation (Subhiksha, S. R, 2026). Without early intervention, these issues can grow and negatively impact a student's long-term academic success.

I chose this problem because schools are now facing increasing challenges while working with limited staff and budgets (Reichart et al., 2026). Teachers, counselors, and administrators are responsible for monitoring hundreds or even thousands of students, which makes it difficult to quickly identify patterns in attendance data. As a former high school science teacher, I experienced this challenge firsthand. While teachers want to support every student, the large number of students makes it difficult to closely monitor attendance trends and identify those who may need help early on.

The goal of using artificial intelligence and machine learning in this scenario is to analyze attendance records and identify patterns that may indicate when a student is at risk. AI models could analyze historical attendance data, behavioral records, academic performance, and other indicators to predict which students may need additional support. Predictive analytics could alert school staff when a student's attendance pattern suggests they may begin missing more school or disengaging from learning.

By providing early warnings and data-driven insights, AI systems could help school staff focus their time and resources on the students who need assistance the most. This could include connecting students with tutoring, counseling, or social services, particularly for those experiencing housing instability or other personal challenges. Ultimately, the goal of this project is to improve student attendance, provide earlier support for struggling students, and increase the likelihood that more students successfully remain in school and graduate.

D.

Linear Regression was chosen for this project because the goal is to predict student attendance rates, which are continuous numerical values. This model fits that type of problem well and produces results that are easy to read and explain. This matters in a school setting, where administrators need to understand what is driving the predictions. Linear Regression also makes it clear how factors like student group, past attendance, and student counts relate to current attendance.

Decision Trees, Random Forest, Logistic Regression, and K-Nearest Neighbors were also reviewed. Decision trees and Random Forest models can capture more complex patterns in the data, but they are harder to explain in a straightforward way. Logistic Regression works better for classification problems, such as predicting whether a student falls below a certain attendance threshold, rather than predicting exact attendance rates. K-Nearest Neighbors is simple, but it can slow down with larger datasets and does not offer much insight into why predictions are made. In the end, Linear Regression is the best fit, because it balances accuracy with clarity, which is important when sharing results with a school board.

E.

The current model relies on a single 80/20 train-test split, which may not provide a fully reliable estimate of performance. Using k-fold cross-validation, like $k=10$, would allow the model to be trained and evaluated on multiple subsets of the data. This approach reduces the likelihood that the reported R^2 value of 0.6447 is influenced by a single split and provides a more stable measure of performance. Increasing the value of k requires more computing power, as the model must be trained multiple times, but it allows for a more thorough evaluation. With additional computing resources, this process can be expanded further, along with testing more model configurations, to improve overall performance.

F.

The performance and efficiency of the model can be optimized by using a confusion matrix to evaluate how well the model makes predictions. A confusion matrix shows the number of true positives, true negatives, false positives, and false negatives, which provides a more detailed view of model performance than overall accuracy alone. By analyzing these values, adjustments can be made to reduce errors, particularly false negatives, which in this context could mean failing to identify students who need support. Thresholds for classification can be adjusted to improve the balance between sensitivity and specificity, depending on the goals of the model. For example, prioritizing the identification of at-risk students may justify accepting a higher number of false positives. This approach allows the model to be tuned for both effectiveness and efficiency, ensuring that it provides useful and actionable insights while minimizing unnecessary interventions.

G.

The use of machine learning in student attendance analysis raises ethical concerns. The model may reflect bias in the data, especially when it includes categories such as homelessness or disability status (Mehrabi et al., 2021). It also relies on sensitive information, which raises questions about privacy and data handling. There is also a risk that school staff rely too heavily on model predictions. While these predictions are useful, they do not capture the full context of a student's situation. For this reason, AI should be used as a support tool, not a replacement for human judgment.

Before proceeding, several concerns in the dataset need to be addressed. The dataset includes sensitive categories such as homelessness status, disability status, and economic background, which require careful handling to protect privacy and prevent misuse (U.S. Department of

Education, 2023). There is also a risk of bias, as these variables may reflect existing inequalities rather than actual student behavior, which could influence the model's predictions (Barocas et al., 2019). In addition, missing values and inconsistencies across reporting years need to be resolved to ensure the data is accurate and comparable. Addressing these issues is necessary to improve the reliability of the model and support fair and responsible use of the results.

To address these concerns, several steps can be taken during data preparation. Sensitive information should be handled carefully by limiting access, removing any unnecessary identifiers, and working with aggregated data when possible (U.S. Department of Education, 2023). This helps protect student privacy while still allowing for meaningful analysis. Data should also be stored securely and used only for its intended purpose. To reduce bias, it is important to review how variables like student groups are used in the model. One approach is to test the model across different groups to check for uneven outcomes (Barocas et al., 2019). If certain groups are consistently predicted as higher risk, adjustments can be made, such as rebalancing the data or reconsidering which features are included. The goal is to avoid reinforcing existing inequalities. Data quality issues should also be addressed before modeling. Missing values can be filled using appropriate methods, such as averages for attendance rates or reasonable defaults for counts. Inconsistent formats across years should be standardized so that all variables are comparable. Finally, the dataset should be reviewed for errors or outliers that could affect results. These steps help ensure the model is trained on accurate and consistent data.

The design of this model raises several ethical considerations related to how predictions are created and used. One concern is fairness. The model includes variables such as student group, which may reflect broader social and economic inequalities. If these variables strongly influence predictions, the model may produce outcomes that disproportionately label certain groups as at risk (Barocas et al., 2019). This can reinforce existing disparities rather than address them. Another concern is transparency. Models like linear regression are easier to interpret than more complex methods, but the results still require explanation. School staff need to understand what factors are influencing predictions and how those predictions should be used. Without clear communication, there is a risk that results will be misunderstood or applied incorrectly.

There is also the issue of accountability. The model produces predictions, but it does not make decisions. Educators and administrators remain responsible for how the information is used. If the model's outputs are treated as final judgments rather than guidance, students could be unfairly labeled or overlooked. Finally, there is a concern about impact. Predictions about attendance can influence how students are perceived and supported. If used carefully, the model can help direct resources to students who need them. If used carelessly, it could contribute to stigma or lower expectations. For this reason, the model should be used to support intervention and support efforts, not to limit opportunities for students.

To improve fairness, the model should be evaluated across different student groups to check for uneven outcomes (Barocas et al., 2019). If certain groups are consistently predicted as higher

risk, the model can be adjusted by rebalancing the training data or limiting the influence of sensitive variables. In some cases, it may be appropriate to remove certain features or use them only for monitoring fairness rather than for prediction. To address transparency, the model outputs should be clearly explained to school staff. Since linear regression is relatively easy to interpret, coefficients can be shared to show how different factors relate to attendance predictions. Providing simple explanations and documentation helps ensure that users understand what the model is doing and its limitations.

To support accountability, the model should be used as a decision support tool rather than a final authority. Schools can establish guidelines that require human review before taking action based on predictions. This ensures that educators consider additional context that the model does not capture. To reduce negative impact, predictions should be used to guide support rather than label students. For example, results can be used to connect students with resources such as tutoring, counseling, or attendance support programs. Avoiding labels and focusing on intervention helps prevent stigma and supports more equitable outcomes.

References

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